

Hardware CWE SIG Meeting Notes

Meeting Minutes

July 17, 2026

Meeting Attendance

Bob Heinemann	Ashrafi Gulam	Rachana Maitra
Steve Christey Coley	Mohammed	Jason Oberg
Andreas Schweiger	Mohan Lal	Irena Bojanova
Parbati K. Manna	Bruce Monroe	Jeremy Lee
William Ferguson	Amitabh Das	Paul Wortman
Mitchell Poplingher	Sohrab Aftabjahani	

Agenda

- RTL Working Group and Other Updates
- Hardware View Inconsistencies
- CWE 5.0

Meeting Notes

RTL Working Group and Other Updates

Content Development Repository

Steve announced a new “related-to-hardware” label in the CWE Content Development Repository. The label allows users to filter submissions that are specifically related to hardware, addressing prior difficulty in locating topic-specific submissions. Steve noted that a similar need has arisen for the AI Working Group.

RTL draft development

Bob reported continued progress on the simulation-synthesis mismatch weakness. Mature draft text was available for the entry name, summary, and extended description. Most other elements had draft text under review, while Related Weaknesses still required development. The working group had produced three demonstrative example drafts,

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including VHDL and Verilog examples; two had received multiple reviews and one had recently entered review. Observed examples from academic work could cite the source material directly while the associated CVE entries remained nonpublic.

Relationship recommendations

Bob confirmed that the relationship recommendations previously presented by Jason Oberg were being tracked in the content development repository and were planned for inclusion in the next CWE release.

Memory-related CWE entries

Jason reopened a prior discussion about memory-related CWE entries that appear in the software view but not the hardware view. Steve said that adding entries to a view is relatively inexpensive from a content perspective, but the hardware view has been community-developed and should not be expanded without community agreement. He also asked members to weigh this work against other priorities, such as improving existing hardware entries. Bob agreed to recirculate the prior meeting materials and revisit the topic at the next SIG meeting. Jason agreed to provide an updated list of candidate entries after Bob circulates the earlier materials.

RTL secure coding guidance

Rachana asked whether the RTL Working Group planned to create general coding rules for avoiding security vulnerabilities. Bob clarified that the current deliverable is the simulation-synthesis mismatch weakness, which is more abstract than coding rules; more specific RTL coding weaknesses could be future work, but CWE does not generally serve as a coding-rule repository. Steve said CWE can represent mappings to external rules and coding guidelines when a concrete public document exists, although he was not aware of public RTL secure coding guidelines. The topic may be explored by the working group.

Hardware View Inconsistencies

Flattening proposal

Bob explained that CWE-1194, Hardware Design, currently mixes flat and hierarchical presentation. The proposed change will present every CWE entry at the same level within its category, consistent with the software view. Categories and category memberships will remain unchanged. Parent-child relationships encoded specifically for CWE-1194 will be removed to create a flat display, while the hierarchical relationships will remain in CWE-1000, Research Concepts, and continue to be visible on individual CWE entry pages. Bob noted that each view maintains its own relationship, which was the technical nuance underlying the change. After this clarification, Jason said the proposal addressed his concern about losing the relationships entirely.

View access and navigation.

Sohrab asked whether users could choose which view to use. Steve demonstrated that users could navigate directly to a view by entering its identifier and showed that CWE-1000 retains hierarchical relationships. Sohrab requested clearer guidance, so visitors understand how to access the research view.

Homepage naming.

Bob observed that the “All Weaknesses” button on the homepage opens CWE-1000, which may be confusing because the site also uses “all weaknesses” in another context. Rachana agreed that the label should be changed. Steve agreed and noted the naming issue as a to-do item. Bob stated that the hardware view flattening was planned for the next release.

CWE 5.0

Preliminary scope and schema

Steve said CWE 5.0 will introduce a new major schema version to provide additional content flexibility and that a draft schema was expected to be released for review. Bob emphasized that the planning topics were preliminary and selected the areas most relevant to hardware.

Extended description reorganization

Bob and Steve described usability findings showing that Extended Descriptions often mix impact language, mitigation guidance, attacker-oriented wording, background information, and detailed technical nuance. This can create redundancy and make entries harder to scan. The planned direction is to move content into purpose-specific elements such as Common Consequences, Potential Mitigations, and Background Details, while considering new locations for deep technical details and attacker-focused information. The goal is a clearer and more concise structure without losing useful content. Steve said the team had not yet made substantive changes to hardware entries beyond grammar fixes and would consult SIG members when questions arise.

Hardware and AI lifecycle support

Bob explained that the Phase field in Modes of Introduction is based largely on a classic software development lifecycle and does not always fit hardware or AI development. The schema may be extended to support terminology used by those communities. He requested hardware community input on the appropriate lifecycle values and cautioned that the examples shown in the slides were illustrative rather than final.

Content coverage

Bob said hardware CWE entries generally have strong coverage, but gaps remain in Detection Methods, Modes of Introduction, References, and minimal VHDL examples. He asked the community to help identify credible hardware-focused references and to contribute content that reflects hardware review, simulation, verification, design, and integration practices.

Explicit CWE dimensions

Bob described plans to explicitly capture the dimension around which a weakness is generalized, such as resources, behavior, property, or technology. He said this information is often already present in prose but is not separately labeled. Making it explicit could improve navigation, taxonomy maintenance, abstraction explanations, gap analysis, hardware-oriented views, root-cause use cases, and AI reasoning.

AI ingestion

Bob said planned Markdown-based outputs for individual CWE entries, and possibly for views, could make the corpus easier for AI systems to ingest. He noted that support for view-level Markdown remained under discussion and that clearer labels with less mixed prose would benefit both human and machine users.

Decisions Made

1. Proceed with a consistently flat presentation for CWE-1194 in the next release. Categories and memberships will remain, while the hierarchical relationships will continue to be represented in Research Concepts View 1000.
2. Include the recommendations previously presented by Jason Oberg in the next release.
3. Reopen the community discussion about adding memory-related CWE entries to the hardware view before any entries are added.
4. Treat the homepage “All Weaknesses” label and related view guidance as a usability to-do item.